

LE PUY LOUDES AP, France

WMO: 074710

Lat: 45.0745N

Lon: 3.7641E

Elev: 832

StdP: 91.72

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.9	-13.1	1.3	-8.5	-10.7	1.7	-6.5	10.7	4.8	9.5	4.4	1.8	240	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	29.6	17.5	27.7	17.1	25.8	16.7	19.1	26.2	18.2	25.1	17.4	24.0	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	13.1	21.8	15.7	12.4	20.9	14.9	11.7	20.1	57.8	26.2	55.0	25.1	52.3	24.0	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.1	7.0	DB	-13.8	32.5	3.1	2.0	-16.0	34.0	-17.9	35.2	-19.6	36.3	-21.8	37.8	(4)
(5)			WB	-14.3	20.6	3.0	0.7	-16.4	21.1	-18.2	21.5	-19.8	21.9	-22.0	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	1.1	1.7	5.0	7.5	11.3	15.4	17.6	17.1	13.3	10.1	4.8	2.0	(6)
(7)		DBStd	6.87	4.05	4.34	3.55	3.33	3.34	3.56	3.11	3.22	3.14	3.76	4.02	4.19	(7)
(8)		HDD10.0	1238	274	232	158	88	25	2	0	0	7	46	159	247	(8)
(9)		HDD18.3	3487	533	466	414	324	218	100	50	63	153	256	406	505	(9)
(10)		CDD10.0	862	0	0	2	14	65	165	237	221	106	48	4	1	(10)
(11)		CDD18.3	69	0	0	0	0	1	13	28	26	2	0	0	0	(11)
(12)		CDH23.3	1069	0	0	0	1	24	213	399	376	53	3	0	0	(12)
(13)		CDH26.7	283	0	0	0	0	2	52	109	115	4	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.7	3.0	3.2	3.2	2.9	2.6	2.6	2.4	2.3	2.4	2.7	2.6	(14)
(15)	Precipitation	PrecAvg	825	50	43	42	71	90	77	75	82	86	82	52	(15)	
(16)		PrecMax	1061	101	94	109	131	159	212	178	190	188	148	197	174	(16)
(17)		PrecMin	601	5	12	10	7	31	20	23	34	21	18	35	12	(17)
(18)		PrecStd	122	24	19	22	32	36	39	34	34	41	36	40	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	15.6	19.0	22.2	26.0	30.9	32.1	32.7	27.2	23.2	16.7	12.6	(19)
(20)			MCWB	8.2	7.8	10.0	13.0	15.3	17.2	17.2	17.0	16.5	14.6	11.8	8.7	(20)
(21)		2%	DB	10.3	12.1	16.2	19.2	23.3	28.1	29.4	29.6	24.9	20.0	14.0	10.8	(21)
(22)			MCWB	7.4	7.0	8.9	11.4	14.8	17.2	17.5	17.5	15.6	14.0	10.8	8.1	(22)
(23)		5%	DB	8.7	9.7	13.9	16.9	21.0	25.8	27.4	27.5	22.7	18.1	12.6	9.5	(23)
(24)			MCWB	6.3	6.3	8.1	10.4	13.7	16.7	17.2	17.2	15.0	13.5	10.1	7.3	(24)
(25)		10%	DB	7.2	7.9	11.6	14.5	18.6	23.6	25.4	25.1	20.4	16.3	11.2	8.3	(25)
(26)			MCWB	5.3	5.3	7.2	9.3	12.5	16.0	16.9	16.8	14.4	12.8	9.1	6.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.1	11.2	13.7	16.9	19.9	20.0	20.0	18.3	16.1	12.5	9.7	(27)
(28)			MCD B	11.7	13.3	16.4	20.6	23.6	27.5	26.5	27.4	23.9	20.2	15.5	11.7	(28)
(29)		2%	WB	7.7	7.6	9.7	12.2	15.3	18.6	19.0	18.9	16.8	14.9	11.3	8.5	(29)
(30)			MCD B	9.8	11.0	14.5	18.1	21.8	25.6	26.1	26.1	22.5	18.6	13.5	10.4	(30)
(31)		5%	WB	6.5	6.6	8.6	10.9	14.2	17.5	18.1	18.0	15.9	13.9	10.2	7.6	(31)
(32)			MCD B	8.4	9.3	13.0	15.8	19.7	23.7	25.2	24.9	21.1	17.3	12.3	9.3	(32)
(33)		10%	WB	5.4	5.4	7.5	9.6	13.1	16.5	17.3	17.1	14.8	13.0	9.2	6.5	(33)
(34)			MCD B	7.1	7.9	11.3	13.7	17.6	22.5	24.2	23.6	19.4	16.0	11.1	8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.5	10.5	10.7	11.1	12.1	12.8	12.9	11.7	9.8	7.9	7.5	(35)
(36)			MCD BR	10.2	12.8	15.5	15.8	15.9	16.5	16.9	17.6	16.6	13.1	10.8	10.1	(36)
(37)			MCWBR	7.4	8.4	9.2	9.1	8.6	7.7	7.1	7.4	8.4	7.7	7.6	7.4	(37)
(38)		5% WB	MCD BR	8.7	11.5	13.3	13.9	14.2	14.6	14.9	15.1	13.8	11.0	9.8	8.6	(38)
(39)			MCWBR	6.6	7.7	8.3	8.2	8.1	7.5	6.9	7.1	7.4	6.8	7.1	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.299	0.331	0.364	0.371	0.374	0.369	0.365	0.343	0.333	0.301	0.278	(40)	
(41)		taud	2.509	2.462	2.407	2.322	2.347	2.366	2.376	2.396	2.442	2.474	2.516	2.520	(41)	
(42)		Ebn at Noon	852	892	903	896	897	893	893	885	878	837	813	815	(42)	
(43)		Edh at Noon	66	83	101	120	121	119	117	110	97	82	66	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.38	2.26	3.45	4.50	5.31	6.28	6.23	5.42	4.17	2.62	1.53	1.20	(44)	
(45)		RadStd	0.17	0.29	0.40	0.56	0.51	0.49	0.49	0.38	0.30	0.27	0.19	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page